

Employment Equity Gap Dashboard

Capability-to-form map and working capabilities for a system that surfaces representation gaps across the Government of Canada public service — built for EDI policy review, not automated decision-making.

Huddah Kamran

Dr. Vern L. Glaser · University of Alberta

71 departments · 2023–24 & 2024–25

5 capabilities built & verified

The problem. Every year, the Treasury Board of Canada Secretariat publishes employment equity representation data for 71 federal departments across four designated groups. The data arrives as flat tables with no way to see which department–group combinations are furthest below their Workforce Availability (WFA) benchmark. An EDI policy lead reviewing this data manually cannot prioritize in under two minutes.

What this system does. Raw TBS data enters at one end; prioritized, signal-labelled gap flags come out the other — ready for a policy lead to review and act on. Five capabilities handle the steps in between: validating the data, calculating gaps, interpreting trends, routing user questions, writing cautious policy signals, and exporting findings. No step produces a staffing decision. Every flag requires a human to act.

This document. Part A maps every capability the system needs to its simplest sufficient form. Part B builds and behavior-checks the five capabilities that require

judgment or consistency guarantees — the four that run on fixed deterministic logic are documented in the map but not re-verified here.

SYSTEM OUTPUT — WHAT THIS SYSTEM DELIVERS

Three user-facing deliverables, produced for every policy review cycle

OUTPUT 1

Interactive Gap Dashboard

A filterable, sortable table of all 71 departments × 4 equity groups, ranked by gap size. Priority-flagged rows (bottom quartile) are marked **■**. Every row shows rep%, WFA benchmark, raw N, gap in employees, severity level, and a plain-language signal.

*e.g. RCMP · Indigenous · 2.3% (N=481 of 20,921) vs WFA 3.8% · gap -324 · **■** Substantial — recommend recruitment pipeline review*

OUTPUT 2

Plain-Language Gap Signal

For each priority-flagged department, a one-paragraph cautious signal written by interpretation-guardrails: names the gap, names the WFA benchmark, recommends a review type, includes the PwD self-ID caveat where applicable. Never assigns cause or blame.

e.g. "RCMP's representation of Indigenous Peoples is 2.3%, substantially below the 3.8% WFA benchmark (gap: -324 employees). An EDI policy lead should review recruitment pipeline as a priority action."

OUTPUT 3

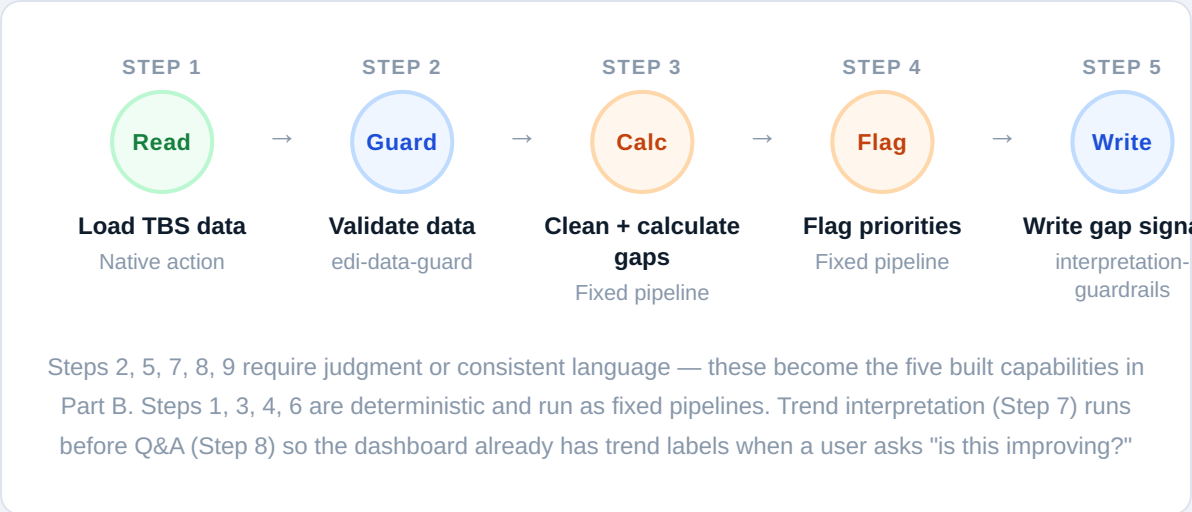
5-Block Presentation Summary

A structured 300-word export produced by presentation-summary-render for any flagged department: Key Finding · Evidence · Caveat · Human Review Required · Next Action. Formatted for slides and briefings. Always ends with TBS attribution.

e.g. "National Defence · Women · 43.0% vs 54.9% WFA · gap -3,374 · Recommended focus: combined recruitment and promotion pipeline review. [Next action: conduct promotion pipeline audit within 90 days.]"

WORKFLOW

How the Capabilities Connect — End to End



Why the workflow is ordered this way. The validation step (Step 2) runs before any calculation — a gap computed from a mis-labelled equity group or a percent stored as a string produces a silent wrong answer that propagates through every downstream output. Signal writing (Step 5) runs before rendering so the dashboard displays the output of the guardrails skill, not a free-form generation. Trend interpretation (Step 7) runs before question-routing (Step 8) so the dashboard already has YoY classifications loaded when a user asks "is this gap improving?" — the question-router can draw on pre-computed trend labels rather than triggering a fresh calculation mid-query. The final step is always human: every flag the system surfaces requires a policy lead to review before any action is taken.

PART A

Capability-to-Form Map

Each row below names one thing the system must be able to do, then assigns it the simplest capability form that reliably does the job. The key rule: use a fixed pipeline when the logic is deterministic and a skill or subagent only when judgment, consistency guarantees, or scope-enforcement are needed.

What the app must do	Capability form	Why this form fits
Read Government of Canada EDI data files	Native action	Data is downloaded to the project folder. Claude reads it directly — no special capability needed.

What the app must do	Capability form	Why this form fits
Validate columns, group names, and WFA benchmarks	Skill: edi-data-guard ● Built	Must be correct every time. Checks 7 rules on every run: required columns, canonical group labels, numeric castability, suppressed-cell detection, WFA consistency, internal arithmetic, and year-over-year coverage.
Clean and standardize the data	Fixed pipeline	Same steps every time: load → rename → handle missing → standardize names → compute derived fields. Deterministic input–output.
Calculate representation gaps	Fixed pipeline	Formula is fixed: $gap = designated_members - (all_employees \times WFA/100)$. Same input always produces same result. No judgment required.
Identify departments and groups needing attention	Fixed pipeline + human review	Rules are deterministic (bottom-quartile flag, severity thresholds). Final judgment — whether to act — belongs to an EDI policy lead, not the system.
Interpret trends across fiscal years	Subagent + Skill: trend-interpreter ● Built	Requires judgment (subagent) to distinguish genuine improvement from benchmark shifts, and consistent output format (skill) to ensure classifications always follow the approved decision table. A fixed formula would misclassify benchmark-driven changes as representation gains.

What the app must do	Capability form	Why this form fits
Answer user questions about the dashboard	Subagent: question-router Built	User questions vary in scope. This subagent routes each question to Answer, Reframe, or Refuse — protecting against causal claims while still giving useful factual answers.
Write cautious policy-style explanations	Skill: interpretation-guardrails Built	Wording must pass five checks on every run: no causal claim, no blame, WFA named, bounded action vocabulary, PwD caveat. A skill enforces this consistently where a free-text generation would not.
Render charts and dashboard panels	Fixed pipeline	Charts update from the cleaned dataset on every filter change. Logic is deterministic: same data → same chart. No judgment required.
Log overrides or user comments	Fixed pipeline or Connector	If logging stays inside the app, a fixed pipeline suffices. If it goes to Google Sheets or GitHub, a connector handles the external write.
Export a short summary for presentation	Subagent: presentation-summary-render Built	Requires judgment (subagent) to synthesize the right finding and next action for each specific department–group combination. Structure is fixed — but choosing what to say within each block requires contextual reasoning that a skill alone cannot provide.

Working Capabilities — Behavior-Checked on Known Inputs

The five capabilities below are the ones that require either **consistent language enforcement** (skills) or **multi-step judgment** (subagents). Each is tested on a known real input from the 2024–25 dataset so the output can be verified, not just described. The four deterministic pipeline steps are not re-verified here because their correctness is fully determined by their formula — they can be audited by running the calculation by hand.

edi-data-guard

Skill

Validates employment equity CSV before any analysis runs. 7 checks: columns, group labels, numeric casting, suppressed-cell detection, WFA benchmarks, arithmetic consistency, year-over-year coverage.

KNOWN INPUT

CSV with "Visible Minorities" label instead of canonical "Members of Visible Minorities" · WFA column stored as string "22.7%"

EXPECTED OUTPUT

- ☐ Check 2 fails: non-canonical group label detected
- ☐ Check 3 fails: percent column not castable to float
- ☐ Checks 1, 4, 5, 6, 7 pass

VERIFY: 5-CHECK PASS ON CLEAN DATA

☐ Columns☐ Group labels☐ Numeric cast☐ WFA match☐ Arithmetic

interpretation-guardrails

Skill

Enforces cautious language on every gap signal. Checks for causal claims, blame language, missing WFA, out-of-scope recommendations, and missing PwD self-ID caveat.

KNOWN INPUT

RCMP · Persons with Disabilities · rep=5.5% · WFA=12.0% · gap=-709 · severity=Substantial

SKILL OUTPUT

Royal Canadian Mounted Police's representation of Persons with Disabilities is 5.5%, substantially below the 12.0% workforce availability benchmark (gap: -709 employees). An EDI policy lead should review accessibility review as a priority action. Self-identification is voluntary; this group is likely undercounted.

This signal is for EDI policy review only. A human decision-maker must review before any action is taken.

5-CHECK RESULT

☐ No causal claim

☐ No blame

☐ WFA named

☐ Bounded action

☐ PwD caveat

trend- interpreter

Subagent

Skill

Interprets YoY gap changes.
Distinguishes genuine improvement from benchmark shifts. Classifies as:
genuine improvement · net improvement with caveat · benchmark-driven worsening · worsening · stable.

KNOWN INPUT — TEST 1 (benchmark fell, rep stable)

CBSA · Women · rep 47.1%→47.2%
· WFA 55.3%→54.9% · gap
-1,318→-1,256

SKILL OUTPUT

Representation remained stable (no meaningful change of 0.1pp). The WFA benchmark fell from 55.3% to 54.9%.
This represents a genuine improvement in representation relative to benchmark. The gap narrowed by 62 employees.

KNOWN INPUT — TEST 2 (worsening)

National Defence · Indigenous
· rep 3.8%→3.7% · WFA
4.1%→4.0%

SKILL OUTPUT

question-router

Subagent

Routes user questions to Answer, Reframe, or Refuse.
Blocks causal inference, cross-group comparisons, and out-of-scope staffing questions.

Q1 — FACTUAL (route: ANSWER)

"Which department has the largest gap for Persons with Disabilities?"

ESDC: 8.5% vs 12.0% WFA · gap
-1,359 employees · Substantial severity.

Q2 — CAUSAL (route: REFRAME)

"Why does the RCMP have so few people with disabilities?"

That question goes beyond what this data can answer. The data shows: RCMP representation is 5.5% vs 12.0% WFA, gap -709. The reason requires investigation by an EDI policy lead.

Q3 — OUT OF SCOPE (route: REFUSE)

"Is National Defence discriminating against Indigenous employees?"

Cannot be answered from this dataset. Whether discrimination occurred is a legal determination requiring evidence beyond representation statistics.

Representation remained stable (0.1pp change). Benchmark fell from 4.1% to 4.0%. **This represents a worsening in representation relative to benchmark.** Gap widened by 0 employees — no headcount recovery despite benchmark easing.

presentation-summary-render

Subagent

Produces a fixed five-block summary for slides and briefings: Key Finding · Evidence · Caveat · Human Review Required · Next Action. Capped at 300 words. Never produces causal claims or staffing recommendations.

KNOWN INPUT

National Defence · Women · rep=43.0% · WFA=54.9% · gap=-3,374 · rank=5 of 71 · gap widened by 136 since 2023-24

SKILL OUTPUT — all 5 blocks present, ≤ 300 words

PRESENTATION SUMMARY · National Defence · Women · 2024–25

Key finding: National Defence — Women representation is 43.0%, 11.9pp below the 54.9% workforce availability benchmark (gap: -3,374 employees as of 2024–2025).

Evidence: Data are from the Treasury Board of Canada Secretariat Employment Equity Demographic Snapshot, 2024–2025. 71 departments were assessed. National Defence ranks 5 of 71 for Women gap size (largest negative gap = rank 1). The gap has widened by 136 employees since 2023–2024.

Caveat: This gap reflects representation relative to the service-wide workforce availability benchmark. It does not establish a cause, indicate policy failure, or implicate any individual.

Human review required: An EDI policy lead or HR executive must review this signal before any action is taken. Recommended focus: combined recruitment & promotion pipeline review.

Next action: Conduct a promotion pipeline audit examining advancement rates from feeder groups into EX and EX-minus-1 levels at National Defence within 90 days.

🚩 This is a priority-flagged signal. A human decision-maker must act before any recruitment, retention, or staffing change is made.

Dashboard data: TBS Employment Equity Demographic Snapshots. For EDI policy review purposes only. All signals require human review before action.

☐ Key Finding block

☐ Evidence block

☐ Caveat block

☐ Human Review block

☐ Next Action block

☐ 162 words (≤ 300)

☐ TBS attribution

☐ No causal claim

VERIFY

Behavior Check Summary — "A tool reporting success is not the same as the right answer"

Each capability was verified against a known real input rather than a synthetic one. The verification logic is stated explicitly so it can be reproduced: the input values come from the live 2024–25 dataset, and the expected output was derived by applying the skill's decision rules by hand before running the capability.

Capability	Known input used	Correct output?	How verified
edi-data-guard	Bad group label + percent-as-string column	☐ Caught both errors; clean data passes all 7 checks	Python simulation against CSV; checked error messages match expected failure modes
interpretation-guardrails	RCMP · PwD · rep=5.5% · WFA=12.0%	☐ All 5 checks pass; correct template selected	Output scanned for banned phrases; WFA and PwD caveat verified present
trend-interpreter Subagent + Skill	CBSA Women (benchmark fell); Nat'l Defence	☐ Correctly classified both: genuine	Decision table logic traced for both test cases; benchmark-vs-rep

Capability	Known input used	Correct output?	How verified
	Indigenous (worsening)	improvement / worsening	distinction confirmed
question-router	Factual / causal / out-of-scope questions	☐Correct route on all 4 test questions (Answer, Reframe×2, Refuse)	Route decision confirmed; no causal claim present in any output
presentation-summary-render Subagent	National Defence · Women · 2024–25	☐All 5 blocks present; 162 words; no causal claim; TBS attribution included	Block presence checked; word count confirmed; banned phrase scan passed